Balls and Strikes: The Effects on Domestic Violence

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Abstract

This paper studies how salient emotional triggers affect domestic violence. Using pitch-by-pitch data from Major League Baseball and crime reports from the National Incident-Based Reporting System, I examine umpire inaccuracies as an exogenous source of emotionally charged frustration. I find that domestic violence reports increase following losses in games officiated by highly inaccurate umpires. The effect is concentrated among the most inaccurate umpires and is driven by missed calls that favor the opposing team. These results are robust across specifications and occur in the hours following a game.

JEL Codes: J12; D91; K42
Key Words: intimate partner violence; domestic abuse; emotional cues; crime

INTRODUCTION

Emotional responses to sporting events have been shown to have real world consequences on familial, judicial, and crime outcomes (Card and Dahl, 2011; Eren and Mocan, 2018; Lindo et al., 2018; Rees and Schnepel, 2009). Notably, Card and Dahl (2011) explored how emotional cues from NFL football game outcomes led to an increase in family violence, finding that upsets during home games led to a 10% increase in domestic violence immediately following the game. This framework has been applied to other areas to measure the effect of emotional cues from sports events on individual behaviors including changes in individual identity, tipping behavior, and charitable donations (Depetris-Chauvin et al., 2020; Ge, 2018; Minnich, 2022).

Sporting events contain numerous potential emotional cues, yet previous research predominantly focuses on the outcome of the game. Additionally, much of the previous work attributes this increase, at least partially, to an increase in alcohol consumption (Trendl et al., 2021; Rees and Schnepel, 2009; Lindo et al., 2018; Ivandić et al., 2024; Klick and MacDonald, 2021). I expand this framework to consider another significant emotional cue mechanism: officiating accuracy. In this work, I extend the existing literature and framework to examine how Major League Baseball (MLB) games, specifically umpire accuracy during these games, can affect domestic violence rates. Specifically, I answer the research question: Does MLB home plate umpire accuracy affect domestic violence rates in the home city of a team? I hypothesize that inaccurate umpiring increases frustration among fans, particularly in the case of calls that disadvantage the team. This heightened emotional response then spills over into increased domestic violence incidents.

To address this question, I acquire pitch-by-pitch data from 2009 to 2019 from baseballsavant.mlb.com, and crime data from the National Incident-Based Reporting System (NIBRS). I focus on reports of male-to-female assault and intimidation committed by spouses, partners, or boyfriends between the hours of 6:00 p.m. and 6:00 a.m., which is when most domestic violence occurs. This provides new insights into the broader implications of emotional cues from sports on societal behavior.

MLB provides an ideal setting for studying the effects of officiating accuracy on emotional responses due to the availability and quality of data on umpire decision-making. Unlike many other sports, where officiating decisions are highly subjective and difficult to quantify, MLB benefits from precise pitch-tracking technology that records the location of every pitch relative to the strike zone. This allows for a somewhat objective measure of umpire accuracy, making it possible to analyze the direct impact of officiating on game outcomes and subsequent emotional responses. Additionally, many MLB broadcasts display a version of the strike zone on-screen, providing fans with a visual reference for judging the fairness of calls, even if this representation may not perfectly align with the official strike zone. Because home plate umpires have significant discretion in calling

balls and strikes, their decisions can substantially influence the flow and outcome of a game, particularly in high-stakes moments. Additionally, MLB offers a uniquely rich dataset due to its long season, with each team playing 162 games per year. This volume of games provides a much larger sample size compared to other professional sports leagues, allowing for more robust statistical analysis and greater variation in umpire accuracy, game outcomes, and emotional responses from fans.

To capture the effect of umpire accuracy on domestic violence, I use a Poisson model with fixed effects for time and space. This allows me to measure how umpire accuracy and the event of a loss affect domestic violence rates within the same time frame and location. I find that umpire inaccuracy increases domestic violence, particularly following a loss. Specifically, less accurate umpires increase the incidence of domestic violence reporting. This effect is strongest for the most inaccurate umpires. I also provide evidence of loss aversion and find the key mechanism for this effect is inaccuracies that favor the other team. My findings are robust to multiple specifications, and I present evidence that this effect is not found for female-to-male domestic violence or other crimes.

While there is a substantial literature on unexpected football losses stemming from Card and Dahl (2011), other research has tried to extend this framework to estimate the effect of sports outcomes outside of football on domestic violence. Kirby et al. (2014) shows a link between soccer outcomes and domestic violence reports in the UK. Similarly, Depetris-Chauvin et al. (2020) find that individuals are more likely to trust others and report reduced violence rates following soccer victories. Munyo and Rossi (2013) also examines soccer games and finds that the outcome of these games affects violent crime in the hour immediately following. Dickson et al. (2016) show increases in domestic violence following rivalry games. Ge (2018) finds that consumers give more generous tips to taxi drivers after unexpected basketball wins, attributing the lack of worse tips after losses to social norms mitigating negative emotional cues while allowing positive ones to influence behavior. Most recently, Cardazzi et al. (2022) finds an increase in domestic violence around unexpected losses in the NBA. They also examine the effect of referee fatigue (and therefore accuracy) as a mechanism for this finding. My work builds on this previous research by examining this potential mechanism driving these effects: officiating rather than just the outcome of the game.

Other research on baseball officiating provides relevant context. Most similar to my current work, Archsmith et al. (2018) estimates the causal relationship between air pollution and umpire accuracy using rich fixed effects and high-quality data, finding that decreases in air quality led to decreases in umpire accuracy. Similarly, Fesselmeyer (2021) finds that umpire accuracy worsens as the temperature increases. Chen et al. (2016) finds evidence that umpires fall victim to the gambler's fallacy, specifically that umpires are more likely to call a ball if the last pitch was a ball and a strike if the last pitch was a strike.

In addition to this, there is a body of literature on how baseball games can have effects on crime in the surrounding area. Mares and Blackburn (2019) shows an increase in various crimes on MLB game days, with these effects increasing as attendance rises, suggesting that viewership may be an important component of these effects. Similarly, Pyun (2019) finds an increase in assaults in Washington, DC when the Nationals, an MLB team, moved from Montreal to Washington, DC. Klick and MacDonald (2021) finds that longer games lead to less crime outcomes, due to stadiums stoppage of alcohol serving following the seventh inning. My study extends this literature by examining the role of umpire accuracy in these crime outcomes.

There are many potential emotional cues happening during sporting events. By incorporating the accuracy of umpiring into the analysis of emotional cues and their impact on domestic violence, my research contributes to a deeper understanding of the mechanisms driving how sports events influence social outcomes. This extension of the literature provides a more comprehensive view of how emotional and environmental factors associated with sporting events can lead to real-world consequences. Understanding the consequences of officiating accuracy has important policy implications, as it suggests that improvements in umpire accuracy may have unintended social benefits by reducing emotionally-driven domestic violence.

DATA

According to the MLB official rulebook, the strike zone is defined as, "the area over home plate from the midpoint between a batter's shoulders and the top of the uniform pants – when the batter is in his stance and prepared to swing at a pitched ball – and a point just below the kneecap. In order to get a strike call, part of the ball must cross over part of home plate while in the aforementioned area. Strikes and balls are called by the home-plate umpire after every pitch has passed the batter, unless the batter makes contact with the baseball (in which case the pitch is automatically a strike)." The home plate umpire makes calls on balls and strikes after every pitch has passed the hitter. If the umpire deems a pitch a strike, it counts as a strike regardless of the spot where it crosses the plate. However, the umpire's judgment can sometimes lead to incorrect calls. This creates a situation where an umpire's accuracy can change a baseball game.

To capture umpire accuracy, I use data from baseballsavant.mlb.com covering all MLB games from 2009 to 2019. MLB tracks an extensive amount of data for each[1] pitch thrown during every game. One key variable provided is the strike zone area, set by an operator using a camera when the ball is halfway from the pitcher's mound to the plate. Although broadcasters provide a similar strike zone for viewers, it is important to note that

[1] Some pitches are not captured in the data because their strike zones are not recorded by the operator. I exclude these pitches from the analysis, meaning they do not count for or against umpire accuracy ratings. These make up about 0.8% of all pitches and appear to be randomly missing.

these are not identical to the official strike zone that I use. For the purposes of this study, I assume that the strike zone reported by Baseball Savant is the correct one and that umpires should use this strike zone when making calls. I classify calls made within the strike zone as strikes and calls made outside the strike zone as balls as correct calls. Conversely, calls made inside the strike zone called as balls and those made outside the strike zone called as strikes are classified as incorrect calls. I then aggregate these calls to the game level and create a variable defined as inaccuracy, which measures the percentage of calls made by the home plate umpire that are not correct in a game. An inaccuracy measure of 10.46 (the mean) means that an umpire called 10.46% of all pitches called in that game incorrectly.

Additionally, there are 288[2] possible states in any inning of a baseball game, depending on the base runners, outs, and the count of balls and strikes. Each of these states has a calculated expected number of runs for a team to score before the end of the inning, based on historical data from the previous five years. Run expectancy is defined as the expected number of runs scored by the batting team before the end of the inning and conditions on the base–out–count state, but not on the current score of the game. Each pitch affects this run expectancy: correct calls change the run expectancy accurately, while incorrect calls change it inaccurately. I measure the effect of umpire accuracy on run expectancy using a variable called Home Favor, which captures the change in run expectancy for the home team due to incorrect calls. For example, a Home Favor of +1 indicates that incorrect calls over the course of the game led to the home team having a one run higher expectancy than they should have had. Conversely, a Home Favor of -0.5 implies that the away team's run expectancy increased by half a run due to umpire inaccuracies. A perfect umpire with 0% inaccuracy would have a Home Favor of zero, but an umpire who misses calls equally for both teams would also have a Home Favor of zero. This is an important possibility, as I use this later to investigate loss aversion by fans.

I also collect data on temperature, attendance, game start-time, game end-time, location, game outcome, and other game-level information from MLB.com. I remove observations from game days where there is a double header. It would be very difficult to discern between the two umpire accuracies on these days. Additionally, I remove any game days that don't have a reported attendance. I supplement this game data with air quality data from the Environmental Protection Agency's Air Quality System. Air quality estimates are captured from the station nearest to the stadium. Specifically, I use the Air Quality Index as a measure of pollution.

For data on crime, I turn to the National Incident-Based Reporting System (NIBRS) to capture family violence for the same 2009-2019 period.[3] NIBRS includes individual-level data on crime reports made to police agencies,

[2] There are 3 possible out states (0 outs, 1 out, 2 outs), 8 possible runner states, and 12 possible balls and strikes states that a hitter may face prior to a pitch being thrown.

[3] Specifically, I use concatenated NIBRS files from Kaplan (2025), which are compiled and cleaned versions of the NIBRS raw data.

including demographics of the victim and offender, and details of the crime reported. The demographics include the relationship of the victim to the offender, and the crime details include the type of crime, time and date, injuries sustained by the victim, and the reporting agency.

I use the NIBRS data to capture domestic violence for each MLB team at the city level, based on the primary police department that services the area surrounding the team's home stadium. Specifically, for each team, I identify the main municipal police department responsible for policing the city in which the team is based and collect all NIBRS reports submitted by that agency. Each reporting agency is identified in NIBRS by a unique Originating Agency Identifier (ORI), which allows reports to be consistently linked to a specific police jurisdiction over time. For example, crime data for the Cincinnati Reds are drawn from reports submitted by the Cincinnati Police Department, while data for the Colorado Rockies come from the Denver Police Department. In some cases, these agencies cover a large geographic area that includes both the city proper and surrounding areas, ensuring that crime occurring near the home stadium of a team is captured. I aggregate these incident-level crime reports to the daily level for each city. The outcome variables therefore measure the number of reported incidents of domestic violence occurring in a given city on a given day. This implies my findings could be a lower bound of the true effect, as games are not only watched by fans near the stadium.

I then merge the aggregated daily crime data to MLB game data at the city–game-day level. The resulting dataset is an unbalanced panel of MLB game days from 2009 to 2019. The panel is unbalanced for several reasons. First, not all police agencies report to NIBRS in all years, and some agencies begin reporting partway through the sample period. Second, I exclude games that are part of doubleheaders, as it is not possible to cleanly assign a single measure of umpire accuracy to these days. Third, I exclude game days for which attendance data are missing; these observations are rare, and the reason for missing attendance is unclear.

Importantly, the final panel includes only MLB game days, not all calendar days. As a result, identification does not rely on comparisons between game days and non-game days. Instead, variation comes from differences across game days within a city, specifically, from variation in umpire accuracy and umpire-induced run expectancy changes across games. Each MLB game day is assigned the number of domestic violence incidents reported in the team's city that night, regardless of whether the team is playing at home or away. Thus, when the Cincinnati Reds play an away game, the analysis still examines how outcomes in Cincinnati change as a function of umpire accuracy in that away game.

Each crime in NIBRS is reported with at least one offense code, and up to ten offense codes. I only look at the first offense code and assume this primary offense captures the entire report[4]. I primarily focus on incidence of simple assault, aggravated assault, and intimidation. However, I supplement this later with reports of rape, larceny, robbery, pocket picking, drug violations, and vandalism or destruction of property.

Similar to Cardazzi et al. (2022) and Card and Dahl (2011), I measure domestic violence as incidents of male-to-female assault or intimidation committed by a spouse, partner, or boyfriend. To capture the effect of domestic violence after the game, I define the day of a game as starting at 6 p.m. and ending at 6 a.m. of the following day, capturing domestic violence that occurs throughout the night. This specification is used because most domestic violence occurs during this time period. Later adjustments to this specification do not significantly impact the results. Additionally, I supplement my results by investigating female-to-male incidence of assault or intimidation. I create this variable in the same way, but flip the genders of the offender and victim.

NIBRS captures reports, not arrests, which is advantageous as it includes cases where no arrest is made. However, one limitation of using this as my measure of domestic violence is that it does not capture victims who do not report the crime. Another limitation is that not all agencies report incidents to NIBRS. However, as long as victim reporting and agency reporting are not correlated with umpire accuracy, this is not a threat to validity.

The low reporting rates to NIBRS restrict the sample of MLB teams available for study, reducing the list of thirty MLB teams to nine[5]. The agencies for the Kansas City Royals (KC), Texas Rangers (TEX), and Minnesota Twins (MIN) began reporting for the entirety of the MLB season in 2010, 2017, and 2019, respectively, so they are not included in years prior to that. My results are robust when excluding them.

Additionally, the Houston Police Department, which services the Houston Astros stadium, begins reporting to NIBRS starting in 2018, but their reporting behavior is much different than all other cities included. Specifically, Houston exhibits an unusual distribution of reported domestic violence incidents, reporting either zero incidents or large spikes (often 10 or more), and very few days with intermediate counts (1–5 incidents). This pattern is not observed in any other city in the sample and is suggestive of reporting irregularities rather than true underlying incidence. As a result, I exclude the Houston Astros from the analysis.

[Table 1 here]

Table 1 shows summary statistics for many of the variables used in my main analysis. The inaccuracy of umpires ranges from 0.74% to 27.27%. This allows me to identify some kind of effect based on this variation of

[4] For the 2009 data, over 92% of all reports in the NIBRS data had only one offense code listed. In 2019, over 91% of all reports only had one offense code.

[5] Cincinnati Reds, Cleveland Guardians, Colorado Rockies, Detroit Tigers, Kansas City Royals, Milwaukee Brewers, Minnesota Twins, Seattle Mariners, and the Texas Rangers

accuracy. The average umpire in this sample calls 10.46% of balls and strikes incorrectly. To better understand this variable, I present a histogram of inaccuracy by team in Figure 1. This shows that most teams see a fairly normal distribution of inaccuracy centered around a mean of about 10.

[Figure 1 here]

The Home Favor variable shows that the average umpire does not change run expectancy in a major way, but there are some cases that umpires can change run expectancy by nearly four and a half runs. These are major changes in the outcome of the game when taken in consideration with the average scores being separated by less than 0.2 runs. Additionally, there is variation in the count of domestic violence cases in a night, ranging from 0 to 21, with the average night in my data having 3.85 reports of domestic violence.

The bottom panel of Table 1 shows domestic violence rates by team city. The Seattle Mariners, whose home stadium is serviced by the Seattle Police Department, have the lowest average domestic violence reports, at just below 1.9 reports of domestic violence per game day. While, the Detroit Tigers, whose home stadium is serviced by the Detroit Police Department, have the highest average domestic violence reports, at
6.49 reports of domestic violence per game day. Additionally, I present histograms for each of these cities in Figure 2. Most locations show a positively skewed distribution, with many days having fewer than 4 reports of domestic violence, but some days that extend past 10 reports.

[Figure 2 here]

METHODS

To assess the impact of umpire accuracy on domestic violence, I employ a Poisson model with fixed effects for space and time. This approach allows me to estimate how an umpire's decisions, particularly incorrect calls, influence domestic violence in the home city of a baseball team. Specifically, I estimate the following Poisson model:

$$E[DV_{st}|Inaccuracy_{st},X_{st},\gamma_{st}] = exp(\lambda Inaccuracy_{st} + \beta X_{st} + \gamma_{st}) \quad (1)$$

where DV_{st} is the raw count of domestic violence incidents reported in city s on day t. The key independent variable, $Inaccuracy_{st}$, measures the percentage of all called pitches that are incorrectly judged by the umpire in city s on day t. The vector X_{st} includes a range of game-specific controls such as temperature, air quality index (AQI), attendance, home and away team records, game end time, and an indicator for whether the game was

played on a holiday. These covariates help mitigate potential biases due to omitted variable concerns by accounting for temporal and environmental factors that could independently influence domestic violence rates.

Fixed effects γ_{st} control for both spatial and temporal heterogeneity. My preferred specification includes city-by-year fixed effects (γ_{sy}), city-by-month fixed effects (γ_{sm}), and city-by-day-of-week fixed effects (γ_{sd}).

The inclusion of city-by-year fixed effects is particularly crucial, as it accounts for time-invariant city-specific characteristics while allowing for differential trends across cities. This absorbs unobserved heterogeneity related to regional law enforcement practices, economic conditions, and other local factors. By incorporating month and day-of-week fixed effects, I control for cyclical patterns in domestic violence, such as seasonal variations and weekend versus weekday reporting dynamics.

I prefer city-by-year fixed effects to separate city and year fixed effects because they allow for differential trends across cities rather than imposing a uniform yearly effect across all locations. Year fixed effects account for common time shocks, and city fixed effects control for time-invariant differences across locations, but together they do not fully absorb local, time-varying shocks. By including city-by-year fixed effects, I control for city-specific trends and allow yearly effects to vary across cities. This approach ensures that local shocks, such as policy changes, economic downturns, or shifts in law enforcement practices, do not bias the estimates. Controlling for these city-level annual shocks is crucial, as they may differentially affect cities in ways correlated with domestic violence.

The same logic extends to seasonal and day-of-week patterns. City-by-month and city-by-day-of-week fixed effects allow these patterns to vary across cities rather than imposing uniform seasonal and daily effects. Domestic violence seasonality likely differs across locations due to varying weather patterns, school calendars, local economic cycles, and holiday timing. Similarly, weekend versus weekday patterns may differ by local drinking culture and work schedules. Since MLB schedules and umpire exposure vary by city, fully interacting these fixed effects ensures that city-specific temporal patterns correlated with both umpire inaccuracies and domestic violence do not bias the estimates.

A central challenge in this analysis is ensuring that umpire accuracy is exogenous to domestic violence outcomes after controlling for observed factors. One concern is that unobservable characteristics associated with game days such as fan aggression, alcohol consumption, or general social unrest may confound the relationship between umpire accuracy and domestic violence. To address this concern, my identification strategy hinges on the assumption that game days with more accurate umpires serve as a valid counterfactual for game days with less accurate umpires. This assumption implies that, conditional on fixed effects and included controls, the variation in umpire inaccuracy is plausibly exogenous. Given that umpires are assigned to games well in advance and that their performance is largely unpredictable ex-ante, this assumption is

reasonable. Additionally, because I only examine game days, and my variation comes from the accuracy of umpires on game days, I avoid conflating the effects of baseball games themselves with the effects of umpire performance.

A related concern is whether domestic violence reporting is systematically different on days with inaccurate umpiring. If, for instance, frustration over incorrect calls leads to an increase in reporting rather than an increase in actual incidents, this could bias my estimates. However, this would require that victims respond to umpire accuracy in a systematic way, which seems unlikely. Additionally, the inclusion of city-by-year, city-by-month, and city-by-day-of-week fixed effects helps control for broader trends in reporting behavior.

Another potential concern is that unobserved factors related to game competitiveness or team-specific characteristics may confound the relationship between umpire accuracy and domestic violence. More competitive games, for instance, may heighten emotional investment and fan aggression, potentially influencing both domestic violence rates and umpire performance. To account for this, I control for both teams' records, which serve as proxies for team strength and fan expectations. Additionally, I control for game attendance, which captures the popularity and perceived importance of a game, as higher attendance could reflect greater fan engagement and emotional stakes. Game end time is another key factor that I control for, as domestic violence incidents tend to be more frequent at night, and longer games may lead to greater fan frustration while also coinciding with natural peaks in domestic violence. These controls help ensure that my estimates capture the impact of umpire inaccuracy itself rather than broader game-related dynamics.

Much of the existing literature on game day effects, such as Lindo et al. (2018), focus on crime outcomes and attribute observed increases to broader, indirect mechanisms, such as alcohol consumption. These studies often rely on telling stories about how game-day dynamics lead to these outcomes, without directly testing the causal pathway. In contrast, my analysis provides a more direct test of the effect of the mechanism (umpire accuracy) on domestic violence by focusing on variation within the game itself. While it is certainly possible that factors like alcohol consumption or heightened emotions could mediate (or enhance) the effect of poor umpiring, my approach isolates the causal effect of umpire inaccuracy on domestic violence, rather than relying on narrative explanations for how game-day dynamics influence these outcomes. By focusing on within-game-day variation, I avoid the need to speculate about the mechanisms and instead offer a clearer, more direct measure of the impact of umpire accuracy on domestic violence.

Finally, I focus much of my analysis on losses, following prior research by Card and Dahl (2011) and Cardazzi et al. (2022), which finds that the emotional effects of losses drive increases in domestic violence. This refinement strengthens my identification strategy by ensuring that comparisons are made within a subset of

games where emotions are more likely to be heightened. By comparing losses with high inaccuracy to losses with low inaccuracy, I further isolate the impact of umpire decisions on domestic violence outcomes.

RESULTS

Using a Poisson regression approach, I find evidence that umpire inaccuracies and losses lead to increases in domestic violence in the home cities of MLB teams. Specifically, when a home plate umpire makes inaccurate calls on balls and strikes, domestic violence rates increase. I find that this effect is driven primarily by the most inaccurate umpires. Additionally, I find that the effect is driven by instances where umpire inaccuracies favor the opposing team in losses. This effect is robust to a variety of timing specifications, and driven by domestic violence that occurs following the game. All regressions control for factors such as temperature, air quality, attendance, team records, game end times, and holidays. Furthermore, all regressions include fixed effects for city, day of the week, month, and year in some combination.

Table 2 presents the main results for equation (1). Column (1) displays the simple relationship between umpire inaccuracy and domestic violence. Column (2) adds controls to the model, while column (3) introduces fixed effects and removes controls. Column (4) includes additive fixed effects along with controls. Column (5) includes day-of-week, month, and city-by-year fixed effects along with controls. Column (6), the preferred specification, includes city-by-year, city-by-month, and city-by-day-of-week fixed effects. Table 2 is divided into three panels. Panel A presents results for all game days in the sample, Panel B focuses on losses, and Panel C includes only wins.

[Table 2 here]

In Panel A, there is no statistically significant relationship between umpire inaccuracy and domestic violence. Turning to Panel B, we observe that umpire inaccuracy has a significant effect on domestic violence in the context of losses. Specifically, in columns (3), (4), (5), and (6), I find a positive relationship between umpire inaccuracy and domestic violence. In the preferred specification (column 6), the coefficient of 0.003 suggests that for each percentage point increase in umpire inaccuracy, domestic violence incidence increases by 0.3%. Using the average inaccuracy rating of umpires in my data, this coefficient implies that a typical umpire in a loss could lead to a little over a 3.1% increase in domestic violence. The most inaccurate umpire, on the other hand, would lead to nearly a 8.2% increase in domestic violence. For comparison, Lindo et al. (2018) find that the incidence of rape increases by 28% on game days, Card and Dahl (2011) report a 10.5% increase in domestic violence following upset losses in the NFL, and Cardazzi et al. (2022) find a nearly 25% increase in domestic

violence in response to upset losses in the NBA on weekends. It is understandable that umpire inaccuracy has a smaller effect than these other sports, as baseball has a much higher frequency of games compared to football and basketball.

The effect I observe is robust across multiple specifications. Panel C, however, shows there is no significant effect of umpire inaccuracy on domestic violence following wins. Taken together, these results indicate that umpire inaccuracy in losses can lead to increased domestic violence, with this effect holding across different specifications. This finding provides evidence of loss aversion, which I explore further later in the paper.

Given that this effect is observed exclusively for losses, I will focus on reporting estimates for the subset of games that end in a loss moving forward, unless otherwise noted.

To further investigate these effects, I analyze umpire inaccuracy by categorizing it into distinct bins. These bins represent different percentile levels of inaccuracy: the bottom 5%, 5%-25%, 25%-50%, 50%-75%, 75%-95%, and above 95%. The most accurate umpires (bottom 5% of inaccuracy) serve as the reference category to prevent multicollinearity. I estimate the relationship between these accuracy bins and domestic violence, restricting the analysis to game days that end in a loss. This regression includes controls and city-by-day-of-week, city-by-month, and city-by-year fixed effects. Figure 3 presents the estimated coefficients for inaccuracy across these bins, along with 95% confidence intervals.

[Figure 3 here]

These results indicate that there is no statistically significant effect for any accuracy bin except the most inaccurate umpires. Umpires in the highest inaccuracy bin (above 95%) have a coefficient estimate of 0.064, which is statistically significant at the 95% level. This finding provides a more complete picture of what is driving the increase in domestic violence. It is not just general umpire inaccuracy but specifically the most inaccurate umpires that lead to these effects. Given the high frequency of baseball games, it makes sense that average umpire inaccuracy would not have a meaningful impact, even in losses. However, when an umpire is especially inaccurate, it may serve as a strong emotional trigger for individuals predisposed to committing domestic violence.

Next, I examine how the umpire's favor within their inaccuracy affects domestic violence rates. This analysis provides additional evidence of loss aversion. Table 3 presents results separately for cases where the umpire's calls favored the team (Favored) and where they favored the opposing team (Not Favored). Recall that the Home Favor variable captures the composite change in run expectancy based on each individual call made by the umpire. On average, this variable is 0.04 runs, but it ranges from as high as 4.49 runs to as low as -3.54 runs. A positive value indicates that the umpire's calls benefited the team (Favored), while a negative value indicates

that the umpire's calls benefited the opponent (Not Favored). I split Table 3 into two panels. Panel A presents the results for losses, while Panel B presents the results for wins. Given that this analysis primarily relates to loss aversion, making this distinction is particularly valuable.

[Table 3 here]

Looking at Panel B, which examines wins, I find no evidence that umpire favor, whether for or against a team, affects domestic violence rates. However, in Panel A, which focuses on losses, the results reveal a significant interaction between umpire favor and inaccuracy. When umpires favor a losing team, their inaccuracies have no effect on domestic violence rates. However, when umpires favor the opposing team in a loss, their inaccuracies lead to a significant increase in domestic violence incidents. This effect is larger than the findings in my main results because the main results are diluting this finding by combining favored and unfavored games. These findings provide further insight into the loss aversion of fans. However, it's not just that a team loses a game, it's the combination of losing and perceiving that the umpire favored the opposing team that appears to drive the increase in domestic violence.

Table 4 combines the main findings from Table 2, Table 3, and Figure 3 into a single triple-interaction framework. I include an indicator variable, Inaccurate, which equals one if the home-plate umpire's inaccuracy in a given game falls in the top quartile of umpire inaccuracies for that season. I also include Disfavored, an indicator for whether umpire calls favor the opposing team, and Loss, which captures whether the home team loses the game. Interacting these three variables isolates the effect of games officiated by one of the most inaccurate umpires, where those inaccuracies favor the opposing team, and the home team ultimately loses.

Columns (1)–(6) progressively add controls and fixed effects, with column (6) representing my preferred specification. Across all specifications that include fixed effects, the coefficient on the triple-interaction term is positive. In the preferred specification, the estimate is 0.080, implying that games officiated by highly inaccurate umpires who disfavor the home team and result in a loss are associated with an approximately 8 percent increase in reported domestic violence incidents relative to baseline game days.

Additionally, the other interaction terms add some further insight into this mechanism. The interaction of Inaccurate and Loss shows marginally significant negative effects, while the interaction of Disfavored and Loss shows no effects. Taken together these imply that an inaccurate or disfavoring umpire and a loss on their own are not enough to increase domestic violence. It's the occurrence of all three where this effect emerges.

[Table 4 here]

Next, I conduct a robustness check around the timing of the domestic violence to validate my findings using this triple interaction model. First, I vary the time frame used to measure domestic violence incidents. Second, I look at only domestic violence that occurs following the hour the game ended and compare that to reports that occur prior to the hour the game started. The MLB reports the exact time the games start and end, but the NIBRS data do not give an exact time of the reports. Table 5 presents these results.

Table 5 presents the results for varying the time frame of domestic violence. The first column replicates my baseline results using the standard 6:00 p.m. to 6:00 a.m. window, which aligns with prior literature and captures the period when most domestic violence incidents occur. The second column expands this window to 4:00 p.m. to 4:00 a.m., while the third and fourth columns narrow the window to 8:00 p.m. to 6:00 a.m. and 10:00 p.m. to 6:00 a.m., respectively. In the fifth column, I use a Post-Game measure, which captures all domestic violence incidents occurring from the hour after the game ends through 6:00 a.m. the following morning. I also use a Pre-Game variable, which captures domestic violence incidents from 7:00 a.m. until the hour the game starts.

[Table 5 here]

Examining these specifications, I find positive and significant effects for the 8:00 p.m. to 6:00 a.m. and 10:00 p.m. to 6:00 a.m. windows, indicating that my results are not overly sensitive to the precise time frame chosen. Additionally, I find that inaccuracies in losses primarily lead to increases in domestic violence during the post-game period, with no evidence of an effect in the pre-game period. This is an important validation check because inaccuracies in losses should not be associated with increases in domestic violence before the game ended. If this were true, it would undermine the causal interpretation of my findings, as the game's outcome would not yet be known. However, finding no evidence of a relationship here supports my causal interpretation. Taken together, the results in Table 5 demonstrate that my findings are robust to different time specifications and that the observed effects are not driven by incidents following games.

To address concerns about game location, Appendix Table A1 estimates the triple interaction model for only home and only away games. I find point estimates that are larger for home games than away games, but neither show significant effects. This is some evidence that these effects are driven mostly by home games.
Additionally, Table A1 shows this triple interaction model for the subset of games that have "Healthy" or "Moderate" air quality index measures and for the subset of games that exclude the hottest temperatures. I find similar effects to the full sample, addressing concerns of endogeneity across these two confounders.

Finally, I turn to investigate other crime outcomes. There is prior evidence suggesting that outcomes of major sporting events can influence a range of behaviors beyond domestic violence, affecting both male and female

offenders and victims. To explore whether similar patterns emerge in my data, I extend the analysis to include all reported crimes, regardless of the gender of the offender or victim, and look specifically at female-to-male domestic violence. The results are summarized in Table 6, broken into two panels.

Panel A of Table 6 presents estimates violent crime including simple assault, intimidation, aggravated assault, rape, larceny, and robbery. Panel B looks at other crimes including pocket-picking, drug violations, and vandalism or destruction of property, and female-to-male domestic violence. As in the previous regressions, all models include fixed effects for the city-by-year, city-by-month, and city-by-day-of-week effects.

Across all of these crimes, I find no evidence of any significant relationship between these outcomes and umpire accuracy. This lack of a relationship is unsurprising given that much of the literature linking such effects to crime partially attributes them to an increase in the number of people in an area. Since I am only examining game days, I do not capture the influx of people that might otherwise be associated with sporting events. Additionally, if crimes are committed due to people attending games, it may be difficult for attendees to discern whether an umpire is calling a game accurately or inaccurately. This suggests that the observed effect is likely driven by male-to-female domestic violence within the home, as these individuals are the ones watching the games and can more clearly observe the umpire's performance.

This result provides further evidence that the observed effects are driven primarily by male viewers, who may be more susceptible to the emotional triggers associated with the game and its outcomes. While it is possible that women could also be influenced by umpire inaccuracies, the absence of a significant relationship in the female-to-male model suggests that the effect is largely concentrated among male viewers, whose emotional responses appear more closely tied to game outcomes. This distinction reinforces the interpretation that the relationship between umpire inaccuracies and domestic violence is predominantly a male-to-female phenomenon.

CONCLUSION

In this study, I use NIBRS data from the home cities of nine Major League Baseball teams, focusing specifically on cases of male-to-female domestic violence, supplemented by pitch-by-pitch data from MLB. I present evidence that domestic violence increases following losses in Major League Baseball games when the umpire is inaccurate. These findings extend the literature on emotional cues and domestic violence by demonstrating that factors beyond game outcomes can influence emotional responses. Using various regression models, I show that these effects are driven by the most inaccurate umpires who favor the opposing team in a loss.

These results indicate that the emotional responses found in other settings may be shaped by specific emotional cues that extend beyond a simple win-loss framework. Here, umpire inaccuracies serve as a particularly salient cue, with higher accuracy mitigating frustration and lower accuracy exacerbating it. This provides broader evidence that perceptions of fairness play a crucial role in shaping emotional responses and subsequent behaviors. This implies that fair officiating has broader social implications than just competitive integrity.

While my work highlights the impact of umpire accuracy on domestic violence, it does not capture all aspects of officiating decisions. My analysis focuses exclusively on the accuracy of home plate umpires in calling balls and strikes. However, umpires make numerous other crucial decisions throughout a game, including safe/out calls at each base, determining whether a batter has swung at a pitch, and issuing ejections to players and coaches. These other forms of officiating discretion may also generate emotional cues similar to those presented in this study, representing a key area for future research. Additionally, my work is restricted to a subset of nine MLB teams, limiting the generalizability of the findings across all franchises and fanbases. Future research could expand to include more teams and a broader range of officiating decisions to provide a more comprehensive understanding of the effects of referee accuracy on emotional responses.

Another promising avenue for future research involves developing methods to quantify particularly salient blown calls. Not all officiating errors carry the same weight. An incorrect strike call in the first inning likely generates a weaker emotional response than a missed call in a high-leverage situation. While my study captures broad trends in umpire accuracy, future work could seek to identify and measure the impact of game-changing blown calls. This is particularly relevant in other sports such as football, where each game carries more significance due to a shorter season. For example, a blown call in the final moments of an NFL game may have a more pronounced emotional impact than a single missed strike call in a 162-game MLB season. Developing a framework to quantify the emotional salience of officiating errors across different sports and contexts could significantly enhance our understanding of the role that perceptions of fairness play in shaping fan behavior and societal outcomes.

An important policy implication of this research concerns Major League Baseball's ongoing efforts to enhance officiating accuracy. Since 2008, MLB has gradually expanded its use of replay review, initially allowing umpires to use video review at their discretion and later implementing manager challenges to correct missed calls. However, these systems currently exclude balls and strikes. MLB has recently begun experimenting with automated strike zones, or "robot umpires," which allow for balls and strikes to be challenged. After testing automated ball-strike (ABS) systems in Minor League Baseball, MLB introduced them in spring training and during the 2025 All-Star Game, and has approved their use in the 2026 season as part of the challenge system.

This development could fundamentally alter the role of home plate umpires. Given the evidence that umpire inaccuracy can exacerbate negative emotional responses among viewers, the adoption of robot umpires may not only improve the integrity of the game but also help reduce the unintended social consequences associated with officiating errors.

While they are implementing robot umpires, this system does not perfectly eliminate the ability of the home plate umpire to still influence the game. My findings align with the work of Bradbury (2019), which shows how umpire performance is particularly sensitive to league directives and monitoring. MLB may consider increasing monitoring of or providing additional guidance to umpires on the implications of accuracy and favor to minimize poor umpiring that could increase domestic violence.

Finally, my findings underscore the broader importance of fair play in professional sports, extending beyond competitive fairness to real-world social consequences. As MLB continues to refine its officiating practices, ensuring greater accuracy and reducing perceived unfairness may not only improve the game itself but also mitigate the unintended negative effects that extend beyond the ballpark.

References

Archsmith, J., Heyes, A., and Saberian, S. (2018). Air Quality and Error Quantity: Pollution and Performance in a High-Skilled, Quality-Focused Occupation. *Journal of the Association of Environmental and Resource Economists*, 5(4):827–863.

Bradbury, J. C. (2019). Monitoring and Employee Shirking: Evidence from MLB Umpires. *Journal of Sports Economics*, 20(6):850–872.

Card, D. and Dahl, G. B. (2011). Family violence and football: The effect of unexpected emotional cues on violent behavior. *The quarterly journal of economics*, 126(1):103–143.

Cardazzi, A., McCannon, B. C., Humphreys, B. R., and Rodriguez, Z. (2022). Emotional cues and violent behavior: Unexpected basketball losses increase incidents of family violence. *The Journal of Law, Economics, and Organization*.

Chen, D. L., Moskowitz, T. J., and Shue, K. (2016). Decision making under the gambler's fallacy: Evidence from asylum judges, loan officers, and baseball umpires. *The Quarterly Journal of Economics*, 131(3):1181– 1242.

Depetris-Chauvin, E., Durante, R., and Campante, F. (2020). Building nations through shared experiences: Evidence from African football. *American Economic Review*, 110(5):1572–1602.

Dickson, A., Jennings, C., and Koop, G. (2016). Domestic violence and football in Glasgow: are reference points relevant? *Oxford Bulletin of Economics and Statistics*, 78(1):1–21.

Eren, O. and Mocan, N. (2018). Emotional judges and unlucky juveniles. *American Economic Journal: Applied Economics*, 10(3):171–205.

Fesselmeyer, E. (2021). The impact of temperature on labor quality: Umpire accuracy in major league baseball. *Southern Economic Journal*, 88(2):545–567.

Ge, Q. (2018). Sports sentiment and tipping behavior. *Journal of Economic Behavior & Organization*, 145:95–113.

Ivandić, R., Kirchmaier, T., Saeidi, Y., and Blas, N. T. (2024). Football, alcohol, and domestic abuse. *Journal of public economics*, 230:105031.

Kaplan, J. (2025). Jacob Kaplan's concatenated files: National incident-based reporting system (NIBRS) data, 1991–2024. ICPSR Data Collection. Ann Arbor, MI: Inter-university Consortium for Political and Social Research [distributor].

Kirby, S., Francis, B., and O'Flaherty, R. (2014). Can the FIFA world cup football (soccer) tournament be associated with an increase in domestic abuse? *Journal of research in crime and delinquency*, 51(3):259– 276.

Klick, J. and MacDonald, J. (2021). Sobering up after the seventh inning: Alcohol and crime around the ballpark. *Journal of quantitative criminology*, 37:813–834.

Lindo, J. M., Siminski, P., and Swensen, I. D. (2018). College party culture and sexual assault. *American Economic Journal: Applied Economics*, 10(1):236–265.

Mares, D. and Blackburn, E. (2019). Major league baseball and crime: Opportunity, spatial patterns, and team rivalry at St. Louis Cardinal games. *Journal of Sports Economics*, 20(7):875–902.

Minnich, A. (2022). Do fans' emotions influence charitable donations? evidence from monetary and returnable cup donations in German soccer stadiums. *Journal of Behavioral and Experimental Economics*, 96:101807.

Munyo, I. and Rossi, M. A. (2013). Frustration, euphoria, and violent crime. *Journal of Economic Behavior & Organization*, 89:136–142.

Pyun, H. (2019). Exploring causal relationship between Major League Baseball games and crime: A synthetic control analysis. *Empirical economics*, 57:365–383.

Rees, D. I. and Schnepel, K. T. (2009). College football games and crime. *Journal of sports Economics*, 10(1):68–87.

Trendl, A., Stewart, N., and Mullett, T. L. (2021). The role of alcohol in the link between national football (soccer) tournaments and domestic abuse-evidence from England. *Social science & medicine*, 268:113457.

TABLES & FIGURES

Table 1: Summary Statistics

	Mean	SD	Median	Min	Max
Inaccuracy	10.46	3.19	10.24	0.74	27.27
Loss	0.51	0.50	1.00	0.00	1.00
Home Favor	0.04	0.74	0.04	−3.54	4.49
Home Score	4.60	3.11	4.00	0.00	22.00
Away Score	4.43	3.16	4.00	0.00	22.00
Temperature	73.15	11.31	74.00	27.00	107.00
AQI	53.09	22.16	48.00	8.00	208.00
Year	–	–	–	2009	2019
Month	–	–	–	3	10
DV Count (6pm-6am)	3.85	2.88	3.00	0.00	21.00
Team	Mean	SD	Median	Min	Max
CIN	2.30	1.98	2.00	0.00	13.00
CLE	5.65	2.69	5.00	0.00	17.00
COL	2.30	1.81	2.00	0.00	12.00
DET	6.49	2.96	6.00	0.00	21.00
KC	4.06	2.16	4.00	0.00	13.00
MIL	4.38	2.97	4.00	0.00	19.00
MIN	2.28	1.65	2.00	0.00	8.00
SEA	1.87	1.45	2.00	0.00	9.00
TEX	2.00	1.52	2.00	0.00	10.00

Notes: This table presents summary statistics. The top panel shows different variables used as measures of treatment or controls. The bottom panel shows domestic violence counts by team. Inaccuracy measures overall home plate umpire inaccuracy in calling balls and strikes. Home Favor measures the change in run expectancy from inaccurate calls aggregated to the game level. AQI is the air quality index.

Table 2: The Effect of Umpire Accuracy on Domestic Violence

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Wins and Losses Combined						
	0.007	0.006	0.002	0.002	0.002	0.001
	(0.011)	(0.010)	(0.003)	(0.003)	(0.001)	(0.001)
N	12296	12296	12296	12296	12296	12296
Panel B: Losses						
Inaccuracy	0.009	0.008	0.004*	0.004**	0.004***	0.003***
	(0.011)	(0.010)	(0.002)	(0.002)	(0.001)	(0.001)
N	6284	6284	6284	6284	6284	6284
Panel C: Wins						
Inaccuracy	0.004	0.003	0.001	0.001	-0.000	0.000
	(0.011)	(0.010)	(0.003)	(0.003)	(0.002)	(0.002)
N	6012	6012	6012	6012	6012	6012
Controls	No	Yes	No	Yes	Yes	Yes
Day-of-Week Fixed Effects	No	No	Yes	Yes	Yes	—
Month Fixed Effects	No	No	Yes	Yes	Yes	—
City Fixed Effects	No	No	Yes	Yes	—	—
Year Fixed Effects	No	No	Yes	Yes	—	—
City-by-Year Fixed Effects	No	No	No	No	Yes	Yes
City-by-Month Fixed Effects	No	No	No	No	No	Yes
City-by-Day-of-Week Fixed Effects	No	No	No	No	No	Yes

Notes: This table presents the relationship between umpire accuracy and male-to-female domestic violence. Controls include variables for temperature, air quality, attendance, team records, game end times, and holidays. Additionally, these regressions include fixed effects for the day of the week, month, city, and year in different combinations. Standard errors are clustered to the city level. *p < .1, **p < .05, ***p < .01

Table 3: Effects Across Favor of Calls

	Favored			Not Favored		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Losses						
Inaccuracy	0.003	0.002	0.001	0.013	0.007*	0.005***
	(0.010)	(0.002)	(0.003)	(0.010)	(0.003)	(0.002)
N	3063	3063	3063	3221	3221	3221
Panel B: Wins						
Inaccuracy	-0.004	-0.000	-0.002	0.011	0.002	-0.000
	(0.009)	(0.004)	(0.003)	(0.012)	(0.004)	(0.003)
N	3066	3066	3065	2946	2946	2946
Controls	No	Yes	No	Yes	Yes	Yes
Day-of-Week Fixed Effects	No	No	Yes	Yes	Yes	—
Month Fixed Effects	No	No	Yes	Yes	Yes	—
City Fixed Effects	No	No	Yes	Yes	—	—
Year Fixed Effects	No	No	Yes	Yes	—	—
City-by-Year Fixed Effects	No	No	No	No	Yes	Yes
City-by-Month Fixed Effects	No	No	No	No	No	Yes
City-by-Day-of-Week Fixed Effects	No	No	No	No	No	Yes

Notes: Favored captures all games where the team is favored by the inaccuracies of the umpire. A full description of how this is calculated can be found in the text. All regressions include controls for temperature, air quality, attendance, team records, game end times, and holidays. Additionally, these regressions include city-by-year, city-by-month, and city-by-day-of-week fixed effects. Standard errors are clustered to the city level. *p < .1, **p < .05, ***p < .01

Table 4: Triple Interaction Model

	(1)	(2)	(3)	(4)	(5)	(6)
Inaccurate	0.006	0.004	0.012	0.013	−0.001	0.002
	(0.021)	(0.021)	(0.017)	(0.017)	(0.011)	(0.011)
Loss	−0.037***	−0.045***	−0.018	−0.025	0.002	0.004
	(0.011)	(0.015)	(0.016)	(0.017)	(0.013)	(0.012)
Disfavored	0.011	0.005	0.016	0.011	0.001	0.002
	(0.019)	(0.016)	(0.017)	(0.017)	(0.016)	(0.015)
Inaccurate × Loss	0.001	0.003	−0.046	−0.046	−0.039*	−0.046*
	(0.027)	(0.026)	(0.028)	(0.028)	(0.023)	(0.024)
Inaccurate × Disfavored	0.034	0.036	−0.006	−0.007	0.001	−0.003
	(0.026)	(0.029)	(0.015)	(0.015)	(0.014)	(0.015)
Loss × Disfavored	−0.025	−0.021	−0.026	−0.020	−0.026	−0.027
	(0.025)	(0.026)	(0.022)	(0.022)	(0.019)	(0.018)
Inaccurate × Loss × Disfavored	0.019	0.014	0.075**	0.073**	0.078***	0.080***
	(0.043)	(0.048)	(0.031)	(0.033)	(0.029)	(0.030)
N	12296	12296	12296	12296	12296	12296
Controls	No	Yes	No	Yes	Yes	Yes
Day-of-Week Fixed Effects	No	No	Yes	Yes	Yes	—
Month Fixed Effects	No	No	Yes	Yes	Yes	—
City Fixed Effects	No	No	Yes	Yes	—	—
Year Fixed Effects	No	No	Yes	Yes	—	—
City-by-Year Fixed Effects	No	No	No	No	Yes	Yes
City-by-Month Fixed Effects	No	No	No	No	No	Yes
City-by-Day-of-Week Fixed Effects	No	No	No	No	No	Yes

Notes: This table presents the relationship between umpire accuracy and domestic violence. Inaccurate refers to umpires whose inaccuracy rating is in the quartile of inaccuracy (most inaccurate). Disfavored is an indicator variable for if the umpire's calls favor the opposing team. A full description of how this is calculated can be found in the text. Controls include variables for temperature, air quality, attendance, team records, game end times, and holidays. Additionally, these regressions include fixed effects for the day of the week, month, city, and year in different combinations. Standard errors are clustered to the city level. *p < .1, **p < .05, ***p < .01

Table 5: Varying Timeframe for Domestic Violence

	6pm - 6am	4pm - 4am	8pm - 6am	10pm - 6am	Post-Game	Pre-Game
Inaccurate × Loss × Disfavored	0.080***	0.072	0.080***	0.134***	0.073**	0.067
	(0.030)	(0.049)	(0.029)	(0.024)	(0.029)	(0.047)
N	12 296	12 296	12 296	12 296	12 296	12 296

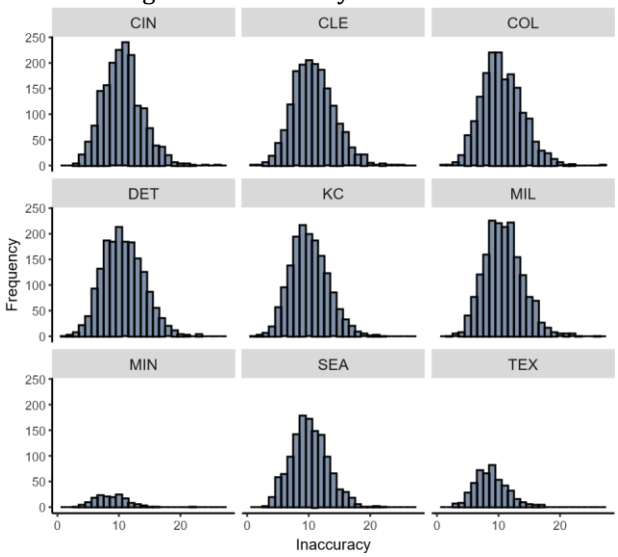
Notes: This table shows the effects of umpire accuracy on domestic violence across various timeframes of domestic violence. 6pm-6am is the main specification I use across all other regressions. Post-Game captures all domestic violence that occurs after the hour the game ends through 6am of the following morning. Pre-Game captures all domestic violence that occurs from 7am through the final hour before the game. All regressions include controls for temperature, air quality, attendance, team records, game end times, and holidays. Additionally, these regressions include city-by-year, city-by-month, and city-by-day-of-week fixed effects. Standard errors are clustered to the city level. *p < .1, **p < .05, ***p < .01

Table 6: Effects on Other Crime

Panel A: Violent Crime					
	Simple Assault	Intimidation	Aggravated Assault	Rape	Robbery
Inaccurate × Loss × Disfavored	−0.004	−0.008	−0.014	0.122	−0.039
	(0.017)	(0.050)	(0.028)	(0.131)	(0.067)
N	12296	12295	12296	12286	12296
Panel B: Other Crime/Female DV					
	Larceny	Pocket Picking	Drugs	Vandalism	Female DV
Inaccurate × Loss × Disfavored	0.011	0.071	−0.066*	0.022	0.016
	(0.046)	(0.327)	(0.036)	(0.046)	(0.086)
N	12296	11481	10544	12296	12290

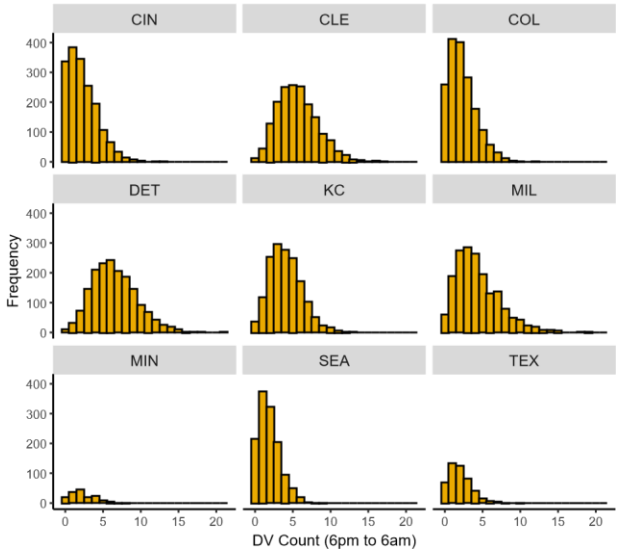
Notes: This table presents the relationship between umpire accuracy and all types of crime. Female DV refers to female-to-male domestic violence. Controls include variables for temperature, air quality, attendance, team records, game end times, and holidays. Additionally, these regressions include city-by-year, city-by-month, and city-by-day-of-week fixed effects. Standard errors are clustered to the city level. *p < .1, **p < .05, ***p < .01

Figure 1: Inaccuracy Distributions



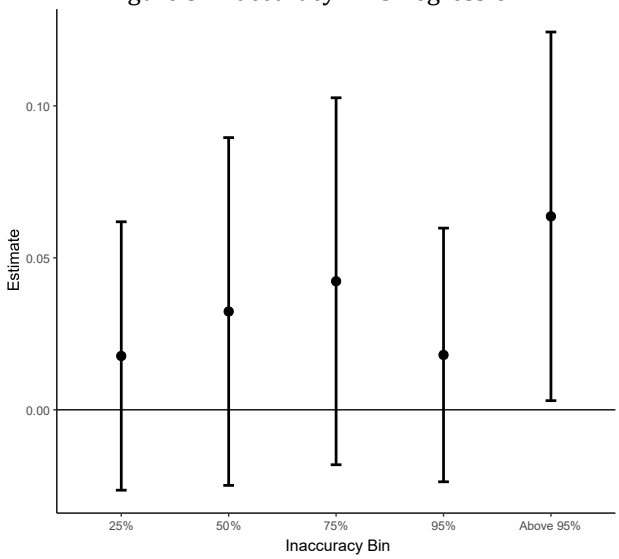
Notes: This figure presents the distribution of umpire inaccuracy for each MLB team in the sample. Inaccuracy is defined as the percentage of incorrectly called balls and strikes by the home plate umpire in a given game. The histogram shows that most teams experience a relatively normal distribution centered around the sample mean of 10.46% inaccuracy, with some variation. Data sourced from Baseball Savant (2009-2019).

Figure 2: Domestic Violence Distributions



Notes: This figure presents the distribution of reported domestic violence incidents (male-to-female assault and intimidation) occurring between 6:00 p.m. and 6:00 a.m. in the home cities of nine Major League Baseball teams. Data is sourced from the National Incident-Based Reporting System and aggregated at the city level. The histograms indicate a positively skewed distribution, with most days experiencing a low number of incidents but some outlier days reporting higher counts.

Figure 3: Inaccuracy Bins Regression



Notes: This figure presents coefficient estimates from a Poisson regression analyzing the effect of umpire inaccuracy on domestic violence. Umpire inaccuracy is divided into bins: the bottom 5% (most accurate umpires), 5–25%, 25–50%, 50–75%, 75–95%, and above 95% (most inaccurate umpires). Point estimates are relative to the most accurate umpires, who serve as the reference category. The results indicate that the observed effect is driven primarily by the most inaccurate umpires. Confidence intervals are at the 95% level.

A APPENDIX

Table A1: Extensions: Game Location and Environmental Context

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Home Games Only						
Inaccurate × Loss × Disfavored	0.060	0.055	0.098	0.097	0.104*	0.100
	(0.076)	(0.084)	(0.067)	(0.076)	(0.063)	(0.070)
N	6136	6136	6136	6136	6136	6136
Panel B: Away Games Only						
Inaccurate × Loss × Disfavored	-0.033	-0.044	0.038	0.030	0.034	0.040
	(0.057)	(0.066)	(0.056)	(0.058)	(0.052)	(0.053)
N	6160	6160	6160	6160	6160	6159
Panel C: Excluding Unhealthy Air Quality						
Inaccurate × Loss × Disfavored	0.040	0.036	0.092**	0.090**	0.087**	0.089**
	(0.052)	(0.059)	(0.038)	(0.040)	(0.034)	(0.035)
N	11778	11778	11778	11778	11778	11778
Panel D: Excluding Hottest Temperatures						
Inaccurate × Loss × Disfavored	0.010	0.003	0.082**	0.084**	0.085***	0.090***
	(0.051)	(0.060)	(0.035)	(0.038)	(0.033)	(0.033)
N	10925	10925	10925	10925	10925	10925
Controls	No	Yes	No	Yes	Yes	Yes
Day-of-Week Fixed Effects	No	No	Yes	Yes	Yes	—
Month Fixed Effects	No	No	Yes	Yes	Yes	—
City Fixed Effects	No	No	Yes	Yes	—	—
Year Fixed Effects	No	No	Yes	Yes	—	—
City-by-Year Fixed Effects	No	No	No	No	Yes	Yes
City-by-Month Fixed Effects	No	No	No	No	No	Yes

Notes: Panel C excludes days with air quality index category of "Unhealthy for Sensitive Groups", "Unhealthy", and "Very Unhealthy." Panel D excludes the top 10% of hottest temperatures. Controls include variables for temperature, air quality, attendance, team records, game end times, and holidays. Additionally, these regressions include fixed effects for the day of the week, month, city, and year in different combinations. Standard errors are clustered to the city level. *p < .1, **p < .05, ***p < .01